CSA1701-ARTIFICIAL INTELLIGENCE LAB EXPERIMENTS

Program 2 : Python Program to Solve 8-Queen Problem

Aim

To develop a Python program that places N queens on an N×N chessboard so that no two queens attack each other.

Algorithm

- Read the number of queens N from the user.
- Create a list board of size N, where board[row] stores the column of the queen in that row.
- Start from row 0 and try every column in the current row.
- A position is safe if no earlier queen is in the same column or on the same diagonal.
- If a position is safe, place the queen and move to the next row.
- If no column is safe in a row, backtrack to the previous row and try its next column.
- When all N rows have a queen, print the board with Q for queens and . for empty squares.
- If no arrangement is possible, report that no solution exists.

Program

```
n = int(input("Enter number of queens: ")) board

= [-1] * n

def safe(row, col):     return all(board[i] != col and
abs(board[i] - col) != row - i                for i in
range(row))

def solve(row):     if row ==
n:         return True     for
col in range(n):         if
safe(row, col):
board[row] = col
if solve(row + 1):
return True              board[row] = -1      return
False  if   solve(0):           for   col   in   board:
print("." * col + "Q" + "." * (n - col - 1)) else:
print("No solution exists")
```

Output

```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
================ RESTART: C:/Users/shaik/OneDrive/Documents/2.py ================
Enter number of queens: 8
Q.......
..Q.....
....Q...
.Q......
...Q....
Q.......
Q.......
Q.......
>>>
```

RESULT

Thus, the 8-Queen Problem was solved successfully using Python.